

Algebra 1: Unit 6, Lesson 8

1. Translate the phrase for the inequality.

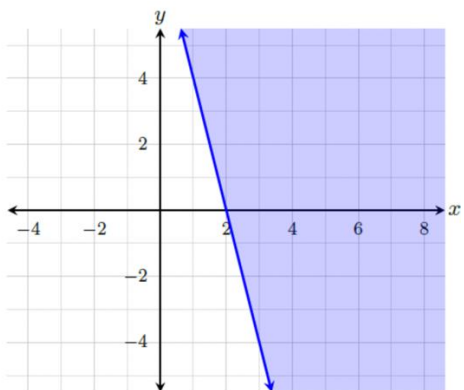
a is at least b

2. Sal is ordering takeout for a party night with 20 guests. A family-sized chicken bucket will feed at least 8 guests and an extra-large chicken bucket will feed 5 guests. This situation can be represented by the inequality $8x + 5y \geq 20$, where x is the number of family-sized chicken buckets and y is the number of extra-large chicken buckets.

The point $(4, -1)$ is not a viable solution because Sal _____ chicken buckets.

3. Josie is shopping for items for a celebration. Value-pack packages of plates contain enough for 100 guests and regular-sized packages contain enough for 25 guests. There must be at least enough plates for 200 guests who plan to attend. This situation can be represented by the inequality $100x + 25y \geq 200$, where x is the number of value-pack packages and y is the number of regular packages.

The combination of value-pack packages and regular packages that will ensure that Josie has at least one plate for each guest is modeled in the graph shown.



Will 1 value-pack and 1 regular pack ensure that Josie has at least one plate for each guest?